Fiscal Year:	FY 2007	**Task Last Updated:**	FY 01/30/2007
PI Name:	Minna, John D. M.D.		
Project Title:	NSCOR: Lung Cancer Pathogenesis and HZE Particle Exposure		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	75390-8593	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2004 NSCOR Space Radiation NNH04ZUU002N
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No. of PhD Candidates:	4	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Start/end dates changed to 3/31/2005-3/31/2010 per grant documents from PI (12/06)		
Key Personnel Changes/Previous PI:	12/2006: Lead of Core D - Bioinformatics and Biostatistics was changed to Dr. Xian-Jin Xie, when Dr. Raul Caetano was named dean of UT Southwestern Allied Health Sciences School and Dr. Harris assumed other responsibilties. Dr. Ruben Ramirez and Dr. Jonathan Abrams are no longer affiliated with this project. Their effort has been absorbed by personnel already supported by the project. They will not be replaced. Also because of his active role in all aspects of the NSCOR leadership, Dr. Jerry Shay has assumed the co-Lead role for the NSCOR Administrative Core for Dr. David Chen.		
COI Name (Institution):	Burma, Sandeep (The University of Texas Southwestern Medical Center) Chen, David (The University of Texas Southwestern Medical Center) Gazdar, Adi (The University of Texas Southwestern Medical Center) Girard, Luc (The University of Texas Southwestern Medical Center) Richardson, James (The University of Texas Southwestern Medical Center) Shay, Jerry (The University of Texas Southwestern Medical Center) Story, Michael (The University of Texas Southwestern Medical Center) Wright, Woody (The University of Texas Southwestern Medical Center) Xie, Xian-Jin (The University of Texas Southwestern Medical Center) Minna, John D. (The University of Texas Southwestern Medical Center)		

Grant/Contract No.:	NNJ05HD36G
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Performance Goal Text:	
Task Description:	The University of Texas Southwestern Medical Center (UTSW) NSCOR proposal focuses on the development of quantitative dose risk estimates following HZE particle irradiation for the development of key genetic, epigenetic, gene expression, and cellular functional changes in the multistep pathogenesis of lung cancer in both new human bronchial epithelial cell (HBEC) and transgenic mouse models of lung cancer. The risk of developing these changes will also be compared in these models to that of X-irradiation. These dose risk assessments will measure events of both radiation-induced promotion (modification of proliferation kinetics of already-initiated cells) as well as radiation-induced initiation (mutational) events. 3D organotypic culture and animal models will also allow measurement and risk estimation of "bystander" effects. These individual risk estimates can then be combined into a model (such as a two-stage clonal expansion model) for overall risk of developing lung cancer from exposure to galactic cosmic radiation. To achieve these goals, UTSW has assembled a team of scientists who are leaders in the study and translation application of the molecular pathogenesis of lung cancer (Minna, Gazdar, Shay); in radiobiology and DNA repair (Chen, Story); a panel of expert Internal and External Advisors (from other NSCORS and lung cancer research experts); as well as expert consultants and collaborators. This team has developed a novel immortalized HBEC system that can be genetically manipulated, studied in 2D monolayer and 3D organotypic cultures. Using these HBECs we have preliminary studies with HZE particle irradiation showing that further progression toward lung cancer can be detected and quantified, that specific expression profiles for HZE irradiation exist, and that HBECs genetically manipulated to progress part of the way toward malignancy are even more sensitive to HZE-induced functional changes. In addition, we have developed plans for novel transgenic mouse models to monitor HZE particle induction of lung cancer in the whole animal in real time. The NSCOR has 4 Projects: 1. Genetic and epigenetic changes in human bronchial epithelial cells following exposure to HZE particle irradiation.; 2. Effect of HZE particles on DNA damage-sensing and repair pathways in human lung epithelial and fibroblast cells; 3. Effect of HZE particle irradiation on functional progression of human lung cancer at the cellular and organotypic level; and 4. Effects of HZE particles on the development of lung cancer in vivo in novel mouse models. These projects are supported by 4 Cores: Administrative; Cell Culture; Expression Profiling and Proteomics; and Biostatistics and Bioinformatics. In addition, with funding from the Department of Energy (DOE) contributing to this NSCOR, studies of low-LET and low dose rate radiation in the same model systems are also being undertaken. The HZE particle irradiation will be done at Brookhaven National Lab, and the biostatistical analysis shows the experiments are powered to help achieve NASA mandated risk estimate confidence levels. Finally, since this group has substantial UTSW institutional commitments, and holds a NCI Special Program of Research Excellence (SPORE) in Lung Cancer and a Center for the NCI Early Detection Research Network (EDRN) program, there will be great synergism and additional resources available for the successful completion of this proposal.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Lung cancer is the leading cause of cancer death in both women and men in the USA and in the Western world. While lung cancer is caused by smoking exposure in 85% of people, because of the large numbers the remaining 15% of cases arising in life time never smokers (~25,000 patients per year) is a sizable health problem in the USA. Likewise, 50% of all new lung cancer cases occur in former smokers (quitting over 5 years before). In all of these scenarios a major underlying question has been the role of other environmental carcinogens especially environmental radiation. One source of this is radon gas in the environment including radon gas exposure in homes with alpha particles (which represent high LET radiation). Thus, the interaction of radiation in the environment on Earth including from alpha particles in the genesis of lung cancer and quantitation of the effect of such radiation on lung epithelial cells is of major health importance. Radiation is a known carcinogenic influence but the molecular events in lung epithelial cells in response to radiation need to be determined. A key element of this is the impact of radiation on lung epithelial cells with preneoplastic lesions. Our NSCOR uses innovative technology of human bronchial epithelial cells (HBECs) developed by us (coming from 36 individuals) which have also been engineered to have known oncogenic preneoplastic changes (such as oncogenic KRAS and knockdown of p53). Our studies are providing quantitative data following high LET irradiation on these HBECs. The primary benefits from this work will be the development of quantitative risk assessment models for astronauts of high LET radiation in space. However, the results will also have major impact and benefit to life on Earth includes new knowledge of the effects on human lung epithelial cells of irradiation in terms of quantitative genetic and epigenetic changes and gene expression changes following irradiation. As part of this will be the quantitative response of DNA repair pathways to radiation including specific DNA repair components such as DNA repair enzymes and signaling pathways in lung epithelial cells. We are also determining quantitative changes in biologic phenotypes of lung epithelial cells in response to radiation which include colony formation in liquid and soft agar (anchorage independent growth) and differentiation and invasion in 3 dimensional cultures, and finally growth in vivo as either xenografts (for human cells) or endogenous lung cancers (for mouse models of lung cancer). From all of the above information we will identify novel molecular biomarkers (mRNA, DNA, protein) for lung carcinogenesis. Finally, we are learning about inter individual variation of the response of HBECs to radiation with respect to the various genetic, epigenetic, mRNA and protein expression, and signaling pathway changes.
	Project 1. Genetic and epigenetic changes in human bronchial epithelial cells following exposure to HZE particle irradiation. Michael Story, Ph.D. Lead, Adi F. Gazdar, M.D. Co-Lead, John D. Minna, M.D. Co-Lead The overall goals of project 1 are to establish the low LET and the HZE radio-response of human bronchial epithelial cells (HBECs) that have been genetically manipulated, with a variety of oncogenic changes that are often seen during the oncogenic progression towards lung carcinogenesis. These changes include: down regulation of p53; oncogenic KRASV12; and over-expression of wild type and mutant EGFR. These cell lines represent in vitro equivalents of preneoplastic lesions and even with multiple changes, these manipulated HBECs show only part of the malignant phenotype and are not capable of forming tumors in immunodeprived mice. We are performing standard survival analysis following radiation exposure, mRNA gene expression analysis, protein expression, DNA methylation, clonogenicity in soft agar (Project 3), and tumorigenicity (Project 3). We follow irradiated cell cultures up to 4 months post-irradiation in order to examine these endpoints in surviving cells in order to determine the effect the radiation exposure has on the oncogenic process. For low LET radiation exposures the radio-response of these cell lines was not

Task Progress:

uniform with mutant EGFR containing cells being radiosensitive. Cells exposed to Fe particles at 1000MeV (1 GeV Fe) are highly susceptible to killing, Si particle exposure at the same energy as the Fe particles, produced an intermediate killing effect, while H exposures showed a similar response as that of gamma-radiation. These results correspond well with the ionization intensity differences seen via DNA foci formation in HZE irradiated cells as determined by Project 2. The cells were collected at various time points following irradiation, the RNA was extracted and Illumina expression microarrays were performed. Acute changes (<24 h) were found in gene expression following exposure of human bronchial epithelial cells to irradiation. These changes were distinct for the various types of irradiation (Fe, Si, gamma) as well as for the various HBEC variants. Most experiments showed no long-term gene expression changes as an effect of irradiation.

Methylation of the promoter region of genes is a fundamental process by which tumor cells inactivate tumor suppressor genes. Our studies have focused on the development of methylation following irradiation of human bronchial epithelial cells (HBECs which are devoid of methylation of multiple genes frequently methylated and silenced in lung cancers. We studied the methylation status of 5 genes frequently methylated in lung cancers – RASSF1A, APC, CYT, GRB38 and RAR-beta. Methylation was measured using a TaqMan semi-quantitative real time PCR method. For all of the unmodified HBEC cell groups and for two of the modified HBECKTR53 cell groups, there were no significant differences between radiated and control groups. However for HBECKRASP53 cell group irradiated with HZE particles, there were important differences where the irradiated group which showed methylation of all 5 genes, and a progressive increase in methylation between the 7 and 14 day time points.

Project 2. Effect of HZE particles on DNA damage-sensing and repair pathways in human lung epithelial and fibroblast cells. David Chen, Ph.D. Project Lead, Sandeep Burma, Ph.D. Co-Lead

With the ultimate objective of estimating cancer risks to humans from HZE particles, in this project we investigate the DNA damage responses of human cells to HZE, the link between these responses and cancer predisposition being very well established. We will take advantage of recent advances in the study of molecular mechanisms of DNA damage sensing, damage signaling, and DNA repair following low LET radiation to verify the hypothesis that HZE particles generate a different and/or unique radiation response in mammalian cells as compared to X-rays. Again we used the panel of isogenic HBEC cells with various oncogenic manipulations described above. Using 1 GeV Fe and X-rays to begin with we have worked out the early damage-responses, damage-repair kinetics, and persistence of DNA damage in primary human skin fibroblasts (HSFs) in great detail. Very briefly, both X-rays and 1 GeV Fe particles elicit similar responses at the sites of DNA breaks. However, Fe-induced damage is repaired more slowly and a large fraction of breaks are not rejoined, thereby resulting in the higher relative biologic effect (RBE) of HZE particles. Using charged particles of increasing molecular weights (O vs. Si vs. Fe) we find that the incidence of unrejoined breaks increases with increasing particle molecular weight presumably due to increasing complexity of the damage induced. Detailed time courses have already been carried out with a panel of HBECs irradiated with 1 GeV Fe particles. The resulting data are currently undergoing analyses.

Project 3. Effect of HZE Particle Irradiation on Functional Progression of Human Lung Cancer. John D. Minna, M.D. Project Lead, Jerry Shay, Ph.D. Co-Lead

Early Events. To evaluate the biologic effects of HZE particle irradiation on HBECs we performed soft agar colony formation assays. Isogenic HBECs with different genetic manipulations were irradiated with HZE and subcultured up to four months after the irradiation. Every month, soft agar colony formation assays were done. The experiments were repeated twice. Most of the oncogenic manipulations reproducibly enhanced the colony formation ability of HBECs without radiation exposure. However, no significant difference was seen between control and HZE-treated HBECs with the one exception of HBECKTR53 (with KRASV12 and p53 knockdown), where HZE irradiated HBECs formed more colonies than non-treated cells at all four time points. This was particularly apparent after correcting for radiation induced decrease in cell survival.

Late Events. We are testing the HZE particle and low LET irradiated HBEC cells of various genotypes for their ability to form tumors in immunodeprived mice as a test of the last events to give full tumor progression. As part of this we are also determining the effect of irradiation on the number of stem cell like cells using new sphere forming assays and quantitative mRNA expression profiling for a panel of ~20 stem cell proliferation program genes.

Project 4. Effects of HZE Particles on the Development of Lung Cancer in Novel Mouse Models. Jerry Shay, Ph.D. Project Lead, James Richardson, DVM, Ph.D.

Transgenic mouse models of human lung cancer have the advantage of a similar tumor development pathway compared to humans but with a shorter developmental period. The LA1 K-ras mouse model we have used contains a latent oncogenic K-ras allele integrated in the endogenous K-ras locus. The activation of the oncogenic allele occurs spontaneously preferentially in the lungs as the animals mature. Every animal containing the latent allele develops alveolar type II lineage lung cancer by the age of four to nine months and all animals die at approximately one year from tumor burden. Our primary objective is to determine if these animals, after low and high LET exposure, initiate or progress to cancer at increased frequency. Our long-term goal is to determine if there is an increase, decrease or no change of onset of death after HZE irradiation and ultimately to use this information for modeling risks in humans. Other end points include other cancer types, DNA damage and repair (gamma-H2AX, DNA-PKc) to correlate with the 2D and 3D cell culture experiments in projects 2 and 3. In addition, we will obtain histology and conduct expression profiles of low and high LET irradiated mouse lungs to compare to the cell culture experiments in Projects 1-3. We have conducted low LET (X-ray) at UT Southwestern Medical Center and high LET (Fe56) at the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory (BNL). 94 animals were irradiated with 0.2 Gy x 5 (1GeV Fe56), 97 animals were irradiated with 1.0 Gy (1GeV Fe56), 110 animals irradiated with 0.2 Gy x 5 (X-ray), and 100 animals irradiated with 1Gy (X-ray). In addition, we have 98 non-irradiated control animals. To examine acute responses (3 hr post fractionated and single dose low and high LET radiation), lung tissue was removed for pathologic examination and RNA isolated for gene expression arrays. Extracted RNA will be hybridized to Sentrix(R) Mouse-6 Expression BeadChips (Illumina). The first section was stained with Hematoxylin and Eosin for histopathological assessment and the following sections are currently being stained with representative markers of the DNA damage response such as gamma-H2AX and 53BP1. Kinetics in normal bronchus, normal alveolus, and tumor tissue will be evaluated separately to determine any differences. We are bringing on line a new mouse model of lung cancer using a transgenic mouse with a Tet-inducible mutant EGFR gene that results in lung cancer. There appears to be a slight decrease in the survival of the irradiated LA1 K-ras mutant and possibly wild-type animals compared to the unirradiated controls. There is also a strain background differences with the B6.1292 background being more

	susceptible to the radiation effects than the 129sv background. So far, irradiated mutant animals do not appear to have increased tumor incidence as determined by post-mortem evaluation. Overall these experiments allow us to study the effects of HZE and low LET radiation on the development of lung cancer in a whole animal model.
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